Question 1

When setting up a monitoring device, how do you choose the sampling rate?

A. Sample as fast as the hardware permits, since a finer time step is always an improvement and allows for better analysis.

B. Make the time step as small as needed for the phenomenon of interest, but as large as possible to spare memory and computation.

C. The time step is imposed by the sensor manufacturer and cannot be chosen.

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Comments

Sampling as fast as possible would be an option if we had unlimited memory and computation capabilities, but it is generally a bad choice.

There is no reason for the manufacturer to constrain the time step. Even if they did so, we can always resample.

The appropriate way to choose a time step is following B.

Question 2

Consider the time series X,

Time (s)	0	1	2	3	4	5
X	0.2	0.3	0.1	0.1	0.5	0.3

Which one(s) of the following time series is a resampling of X ?

A.

Time (s)	0	2	4
Y	0.2	0.1	0.5

B.

Time (s)	0	2	4
Y	0.25	0.1	0.4

C.

Time (s)	0	2	4
Y	0.3	0.1	0.3

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Comments

Each of the proposed time series is a resampling of X, the diffrence lies in the aggregation method. For A, the first time step of the aggregated interval is taken. For B, the average of the aggregated interval is taken. For C, the last time step of the interval is taken.

Question 3

What is the strength of the mean compared to the median?

A. Discards outliers.

B. More robust to missing data.

C. Easy to compute.

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Comments

The mean is particularly sensitive to extreme values, so outliers are not discarded. the mean is neither more or less robust to missing data. It is however easy to compute and to understand.

Question 4

A colleague computes covariances between several pairs of time series from an experiment (temperature, pH, oxygen concentration, aeration rate) and finds the numbers impossible to compare with each other.

Why does switching to correlation resolve this?

A. Correlation detects non-linear relationships that covariance misses.

B. Correlation ignores the sign of the relationship, which covariance does not.

C. Correlation is computed on fewer data points, so it is less noisy.

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Comments

Correlation detect the same properties as covariance, it does not see anything else. Correlation is sensitive to the sign of the relation between time series. Correlation uses exactly the same data points as covariance.

Question 5

When computing a moving average, what should you keep in mind?

A. The time series get longer after processing.

B. The general trend of the time series disappears in the process.

C. The time series' average value is almost unchanged.

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Comments

When applying a moving average, the time series gets shorter. The general trend is actually what is preserved by the moving average, small variations are lost. The average value remains very close to the original one as the general behavior is the same.

Question 6

Using a portable device, you have measured the heart rate of a patient every 10 minutes over a period of 60 days. This gives you a time series $X = (X_0, X_1, X_2, ..., X_{8639})$, where the value X_t is the patient's average heart rate between time t and $t + 5$ minutes.

Computing the ACF (Autocorrelation function) of X, what do you expect to observe?

A. A peak every minute, corresponding to the reset of the measuring device.

B. A peak every ten minutes, corresponding to the recording of the new measurement.

C. A peak every six hours approximately, corresponding to the physiological change of digestion (the patient have their meal at 7h, 13h, and 19h).

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Comments

The time step of the time series is 10 minutes, therefore we cannot compute ACF with a time lag below 10 minutes. Each time lag is 10 minutes apart, therefore it makes no sense to have a peak every 10 minutes. Physiological processes impact heart rate, in particular digestion.